



Viewpoint: Ensuring affordability of diets in the face of shocks

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ABSTRACT

Climate-related shocks, the COVID-19 pandemic, and the war in Ukraine have exacerbated world hunger, making it more difficult for people to meet their nutritional needs. This viewpoint examines, for 136 countries, the ability of populations to afford an energy-sufficient and a healthy diet in the face of shocks. Using pre-pandemic data from 2019, we calculate the percentage of countries' populations at risk of losing access to these diets in the event of a one-tenth, one-fifth, or one-third reduction in real income. We find that, in addition to the 152 million people who were unable to afford a basic energy-sufficient diet in 2019, up to 260 million people (mostly in low-income and lower-middle income countries) are vulnerable to not being able to afford it should a shock reduce real income by up to one-third. The more expensive healthy diet, which was already out of reach for 3 billion people before the pandemic, risks becoming inaccessible to an additional 968 million people (nearly all in middle-income countries). Our findings indicate that addressing income inequality is key when trying to ensure access to energy-sufficient diets, while raising mean income and reducing diet cost are increasingly relevant for ensuring access to healthy diets, especially following a shock. We thus provide insights on how countries facing shocks will need a varying mix of social protection, income stabilization, lowering the cost of nutritious foods, and investing in broader economic development.

1. Introduction

The debate on food security and nutrition has substantially evolved in the last 50 years, shifting from a concern about famines and agricultural supply in the 1970s to a more diversified set of concerns, exemplified by recent work on the affordability of a healthy diet (Bai et al., 2021; Hirvonen et al., 2020; Masters et al., 2018; Raghunathan et al., 2021). Despite the new focus on different forms of malnutrition, eliminating hunger remains a major challenge. Efforts to defeat world hunger were proving insufficient even before the COVID-19 pandemic, with an estimated 690 million people in the world (or 8.9% of the world population) still facing hunger in 2019 (FAO et al., 2020). The COVID-19 pandemic, and the policy responses to contain it, have made the situation even more challenging, contributing to an estimated increase of those affected by hunger of as much as 138 million people (FAO et al., 2022). Lockdowns reduced access to employment and livelihoods, leading to reductions in food spending and, consequently, in food security and nutrition, especially of the poor and vulnerable (see review by Picchioni et al., 2021). More recently, the war in Ukraine has created further pressure on agrifood systems by increasing the (already elevated) prices of food and inputs, further worsening world hunger, food insecurity and malnutrition (Benton et al., 2022; Meijl et al., 2022). FAO's food price index mirrored the impact of these shocks as it

increased by approximately 35% in real terms between December 2019 and July 2022 (FAO, 2022).

In these circumstances, it is vital that countries build resilience in the face of economic adversity to guarantee economic access to food. This is particularly true for the more expensive nutritious foods that make up a healthy diet (Bai et al., 2022), considering that almost 3.1 billion people could not afford a healthy diet in 2020 (FAO et al., 2022). In more extreme cases, the cost of a healthy diet can surpass that of income available for food, thus making the diet inaccessible to populations, even where income is well distributed. In this case, countries should prioritize a combination of increasing incomes and lowering the cost of nutritious foods. In many countries, persistent and high levels of inequality further constrain people's economic access to nutritious foods, which calls for another set of policies and investments that focus on reducing structural inequalities (World Bank, 2022a).

Where even staple foods that make up a basic energy-sufficient diet are out of reach to many, countries may prioritize, in the short-term, ensuring basic nutritional needs. This is challenging given that hunger and undernutrition are often the result of prolonged conflicts and weak institutional capacities (FAO et al., 2017), and addressing limited access to even staple foods will need humanitarian and conflict resolution innovations that are not specific to agrifood systems (Barrett, 2021). The distinction between food security and nutrition challenges that originate

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within agrifood systems and those that are broader is at the center of this viewpoint.

Timmer (2000) outlined the “macro” dimensions of food security, focusing on the aggregate conditions needed to ensure food security, and concluded that policymakers should concentrate on a combination of three approaches: income increase (without redistribution), income distribution (without changing average income per capita) and economy stabilization. Ensuring economy stabilization is especially critical where income is not well distributed since economic downturns are found to disproportionately undermine food security and nutrition where inequalities are greater (FAO et al., 2019). This prescription is relevant to today’s debate on diet affordability, and can be supplemented by the recent work on food system resilience (Tendall et al., 2015) and social protection (Daidone et al., 2019; Devereux, 2016, 2016; Gentilini et al., 2022a; Timmer, 2000; Tirivayi et al., 2016).

Relative to recent commentaries on the (ongoing) global food crisis (e.g. Osendarp et al. (2022) and Pörtner et al. (2022)), this viewpoint goes beyond general prescriptions and dives deeper into the specific challenges of economic access to food. It builds upon the recent work on affordability of least-cost diets (Bai et al., 2021; FAO et al., 2020; Herforth et al., 2020; Hirvonen et al., 2020; Masters et al., 2018) and calculates for 136 countries, with diet costs and income distribution data available for 2019, how many people are unable to afford an energy-sufficient and a healthy diet should a shock affect real incomes. We then decompose the effects of poverty, income inequality and the cost of diets in terms of their statistical significance and relative magnitude in determining the share of population who cannot afford – or is at risk of not affording – an energy-sufficient diet and a healthy diet.

The viewpoint applies the intuition in Timmer (2000) that “rather than asking how to cope with hunger and famine, the question might be how to escape their threat altogether”, and suggests possible pathways in the context of access to diets, including those considered the healthiest. It acknowledges that past price stabilization – mostly referred to single basic staples (e.g. rice in Asia) – may still be relevant for providing access to energy-sufficient diets today, alongside social protection and redistributive policies. However, reducing the cost of a diverse set of nutritious foods will also play a central role in ensuring economic access to a healthy diet, as will a shift towards raising and stabilizing incomes more generally.

2. Quantifying the peril of unaffordability of energy-sufficient and healthy diets

Having economic access to food is an issue related to prices and income, and is therefore shaped by poverty, income inequality and the cost of food relative to disposable income. The poorer a person is, the larger the proportion of income spent on food. Therefore, even small increases in the cost of a diet or small reductions in the level of incomes can have significant impacts on the accessibility to diets by the poorest. Not having a reliable and substantial income buffer limits consumers’ options during periods of shocks, such as price spikes, crop failures, or loss of assets. Our measure of economic access estimates to what extent specific diets are actually accessible to populations and the extent to which this accessibility may worsen in the face of a shock.

2.1. Conceptual framework

To represent the full spectrum of economic access to food, we examine the ability of populations to afford both a basic energy-sufficient diet meeting calorie needs and a healthy diet that protects against malnutrition in all its forms by containing a balanced, diverse and appropriate selection of nutritious foods. In an energy-sufficient diet, 100% of the energy requirement is derived from starchy staples

through the consumption of the cheapest cereal available in the market. Lack of access to it is used here as an indication of extreme deprivation in access to nutritious foods. As expected, the cost of a diet increases as the diet quality increases. The cost of a healthy diet is, on average, almost 5 times the cost of the energy-sufficient diet, and this pattern holds across all regions and country income groups (FAO et al., 2021).

The number of people able to afford a given diet in a given country depends on the diet cost relative to income and on the income distribution of its population. In case a systemic or idiosyncratic shock occurs and causes a reduction in an individual’s real income, the ability to afford the diet will depend on how much of an income buffer that individual has. In the case of a systemic shock (e.g. spike in inflation) the buffer is needed to cover the increased cost of both food and non-food items. In the case of an idiosyncratic shock (e.g. partial loss in income due to illness or unemployment) the buffer is needed to maintain basic expenses, namely food. Fig. 1 illustrates the full spectrum of affordability estimated by this viewpoint. It aggregates across individuals, showing the distribution of disposable income, and assumes a relative cost for each diet. The income buffer is expressed as a given share (x) of disposable income needed to ensure economic access to a given diet.

Starting from the left tail of Fig. 1, the dark red area expresses the share of population who cannot afford an energy-sufficient diet even in the absence of a shock, and thus represents the most vulnerable population in a country. Should a shock affect real incomes (either through a price increase or income shortfall), the share of those unable to afford increases to the end of the light red area. These people lack the needed buffer (x) to guarantee access in the face of a shock and are therefore vulnerable to losing access. The orange area represents the share of population who is resilient to shocks affecting the cost of an energy-sufficient diet, but are still unable to afford the far more expensive healthy diet. Populations in yellow and green areas can afford a healthy diet; however, in case of a shock, those in the yellow area risk losing access to the healthy diet. In sum, the yellow and light red areas indicate populations lacking enough of an income buffer – following a shock – to afford an energy-sufficient diet and a healthy diet, respectively. Having this income buffer is an important strategy for protecting against shocks and stresses, and will therefore be a central concept in this viewpoint. Finally, the green area expresses the share of people who have enough of an income buffer to continue to be able to afford a healthy diet even after a shock.

The alternative income distribution in Fig. 1 (dashed) represents a more equitable distribution for the same mean income. It shows that, when income is more evenly distributed, affordability generally increases. Under the more equitable distribution, everyone can afford the energy-sufficient diet and many more become able to afford a healthy diet. This shows that, for a given mean income and diet cost, a more equitable income distribution plays a critical role in improving affordability; however, while the number of people who cannot afford a diet decreases, the number of people who are vulnerable to not affording it may be unchanged or even increase. Although this is unambiguously a better outcome since more people can now afford the diet under normal circumstances, vulnerability to shocks remains. Recalling Timmer’s objective of “escaping the threat altogether”, ensuring diet affordability – especially a healthy one – will require a higher average income and/or reducing diet cost.

2.2. Data and methods

Based on data for 143 countries for which costs of diets were available, FAO et al. (2021) report that in 2019 there were around 3 billion people who could not afford a least-cost healthy diet. In Fig. 1, these people fall to the left of the yellow area. Out of the 3 billion, approximately 177 million could not meet basic dietary energy requirements

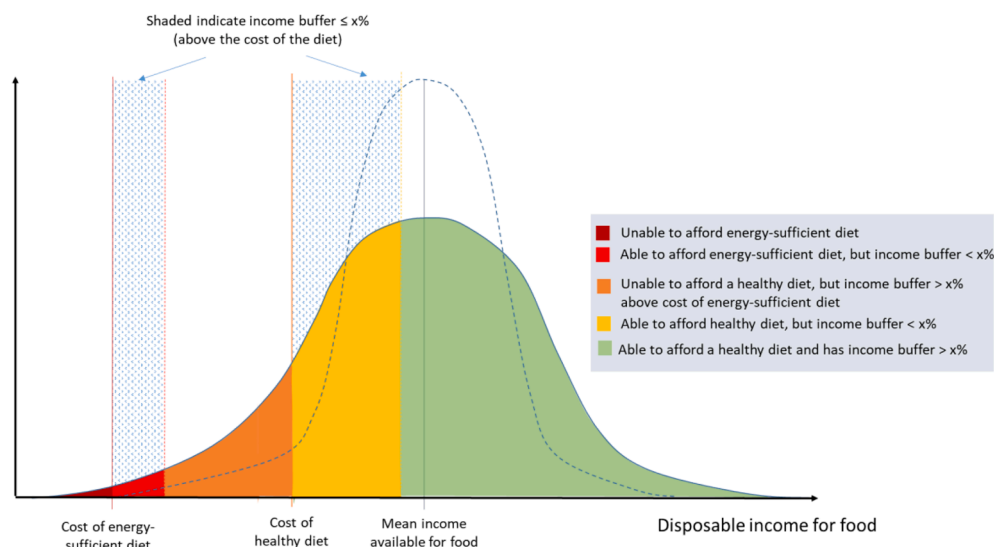


Fig. 1. Representation of the spectrum of affordability and vulnerability to shocks along the distribution of disposable income, considering diet costs.

(represented by the dark red area) (FAO, 2021). This viewpoint supplements this information by calculating, for 136 countries,^a the share of population vulnerable to losing economic access to least-cost diets (energy-sufficient or healthy diets) because they lack a sufficient income buffer to afford them in case their real incomes are reduced. Looking at Fig. 1, these populations are located in the light red and yellow areas for energy-sufficient and healthy diets, respectively.

Diet costs refer to the year of 2019 – the year before the outbreak of the COVID-19 pandemic – and are taken from FAO et al. (2021).^b Because people do not spend all of their income on food, we assume that 37% of expenditures must be reserved for non-food expenditures such as housing, transport, education and health (INDDEx Project, 2022; Lele et al., 2016; Moltedo et al., 2014; Smith and Subandoro, 2007). This is based on observations that the poorest people in low-income countries spend, on average, 63% of their incomes on food (World Bank, 2020). We thus define an *affordability threshold* for a diet equal to (cost of the diet)/0.63, which reflects the income below which people will not be able to afford the diet even if they prioritize food. The percentage of people who cannot afford a diet is obtained based on the affordability threshold and comparing it with 2019 country income distributions from the World Bank Poverty and Inequality Platform (World Bank, 2022b). Percentages are then multiplied by the 2019 population in each country (United Nations, 2019) to obtain the number of people who are not able to afford a certain diet.

To calculate the number of people at risk of losing access to a diet (light red and yellow areas in Fig. 1), we first define the income buffer needed to absorb a given shock – should this shock occur, those without this extra income are at risk of losing access to a given diet. To capture resilience to shocks of varying degrees, we consider three levels of income buffer relative to the affordability threshold for a diet: 11%, 25% and 50%. These buffer levels can be interpreted in two equivalent ways. First, as the extra income needed to continue to be able to afford a diet

should all prices (both food and non-food) increase by 11%, 25% and 50%, respectively. Alternatively, they can be interpreted as having real incomes be reduced by one-tenth, one-fifth and one-third, respectively.^c In other words, when the cost of the diet plus the buffer exceed 63% of a person's income (i.e. the part of income typically spent on food by the poorest in low-income countries), a person is at risk of losing access to that diet if a shock occurs. Therefore, while the threshold to calculate the inability to afford a diet in 2019 is (cost of the diet/0.63) as indicated earlier, the threshold to calculate those at risk is obtained as $(1 + \text{buffer}) \times (\text{cost of the diet}/0.63)$, whereby the buffer can take the values of 11%, 25% or 50%. As before, we compare these thresholds to 2019 country income distributions to obtain the shares of people who are at risk of not affording the diets.

These three scenarios are synthetic and hypothetical and thus not meant to simulate actual changes in real income driven by existing shocks. Nonetheless, the choice of these income buffer ranges is based on recent evidence of the impact of the COVID-19 pandemic and the war in Ukraine on real incomes. In Cambodia, for example, out of the households that were affected by the pandemic, more than 60% suffered an income loss of one-third or more, while 92% suffered one-fifth or more (World Bank, 2021). Another study found that income loss as a consequence of the pandemic was common across the nine countries included in the study, with 51% of respondents in Myanmar and 79% in Liberia reporting their income “reduced a lot” or “completely stopped” (BRAC International, 2020). Another study on Brazilian doctors found that, of those whose income was affected, 31% experienced a reduction of 25–50% and about 60%, a reduction of over 50% (Gomes et al., 2020). In addition, since the beginning of 2022, inflation has become a global phenomenon, due to energy price spikes driven by the Ukraine war. For example, the FAO's food price index shows an increase of approximately 35% between December 2019 and July 2022. In terms of retail prices, Bai et al. (2022) report a 25% increase in the global average food consumer price index (CPI) between January 2019 and June 2021, with considerable variation across countries.

^a 7 countries previously included in FAO et al. (2021) were dropped from the analysis in this viewpoint due to methodological issues later encountered. These countries are Argentina, Comoros, El Salvador, Liberia, Palestine, Togo and Zimbabwe.

^b For a full description of the cost methodology and data sources, see Annex 3 of FAO et al. (2020). Note that this methodology has been updated since then for the healthy diet, and thus improved healthy diet costs have been calculated and published in FAOSTAT (FAO, 2023). However, because the methodology of the energy-sufficient diets remained unchanged, this viewpoint uses relies on the previous methodology and diet costs to ensure consistency between the two.

^c Mathematically, the buffer required to remain able to afford a diet is the reciprocal of the reduced real income following a shock. For example, if a shock reduced real income by one-third (1/3), the shocked individual would have only two-thirds (2/3) of income left, the reciprocal of which is 3/2, or 1.5. This means income needs to be 50% higher for the person to absorb the shock and continue to afford the diet.

3. Results

3.1. Distribution of the global population across a spectrum of diet affordability

Based on the conceptual framework in Fig. 1, Fig. 2 represents the spectrum of affordability of the two diets by country income group, with and without a shock requiring a 50% income buffer (see [Supplementary Information](#) for income buffers of 11% and 25%). Globally, we estimate that, on top of the 152 million people who were unable to meet basic dietary energy requirements before the pandemic, an additional 260 million are at risk of not being able to afford as well if their incomes fall by one-third. These vulnerable populations are concentrated in low-income countries: almost 40 million people (or 8%) are unable to afford the energy-sufficient diet, and an additional 59 million people (or 12%) risk losing access in the face of a shock. Further alarming is the fact that, in lower-middle income countries, 6% of people (162 million) are also at risk, on top of the 3% who already cannot afford it.

As for the healthy diet, we estimate that, on top of the 3 billion people that were unable to afford a healthy diet before the pandemic, up to an additional 968 million people are vulnerable to not affording it.

As expected, only a very small share of the population in high-income countries (1%) is at risk of not being able to afford a healthy diet. This is an expression of Engel's law, whereby consumers in high-income countries allocate a much higher share of income to non-food

expenditures, which can potentially be used as buffer when food prices increase substantially or if they face a shortfall in income. Middle-income countries have made gains in access to healthy diets; however, their population does not have a comparable income buffer, so they are at great risk of seeing a reversal given recent price trends and income shocks. About 18% of the population in upper-middle income countries (474 million people) is at risk of not affording a healthy diet, on top of the 22% who already cannot afford it. The situation is even more dramatic in lower-middle income countries, where 70% already cannot afford a healthy diet and an additional 16% are at risk.

Fig. 3 represents this spectrum by world region. Results somewhat mirror those of Fig. 2:

- (i) In sub-Saharan Africa, there are almost as many people at risk of not affording the energy-sufficient diet as there are unable to afford (12%), equivalent to a total of 25% of the region's population (more than 250 million people).
- (ii) Most people in Southern Asia (99.6%) are able to afford an energy-sufficient diet, although 4% do not have enough income to buffer a shock reducing real incomes by one-third. Furthermore, more than two-thirds were not able to afford a healthy diet in 2019, and out of those who could afford it, more than 50% are at risk (300 million).
- (iii) In Central, Eastern and South-eastern Asia, affordability of energy-sufficient diets is not an issue for most people; however,

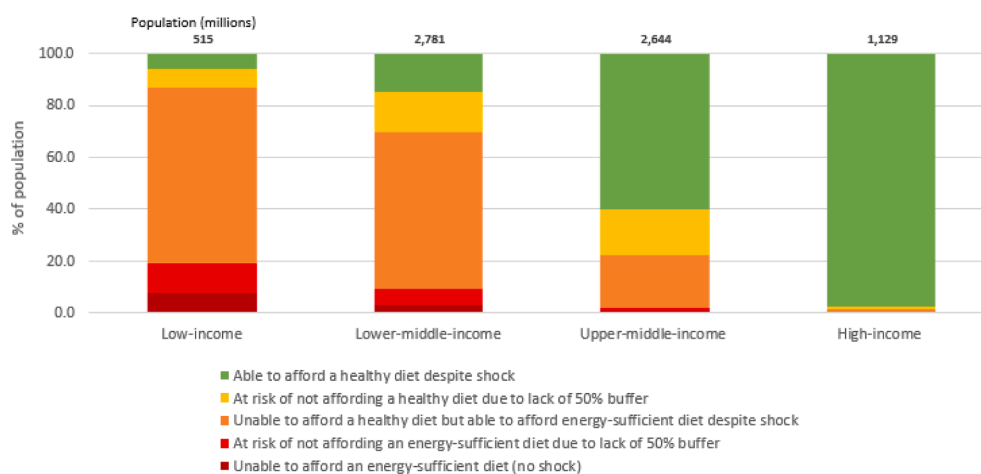


Fig. 2. Spectrum of diet affordability and vulnerability to a shock equivalent to a one-third loss in real income: percent of population, by income group.

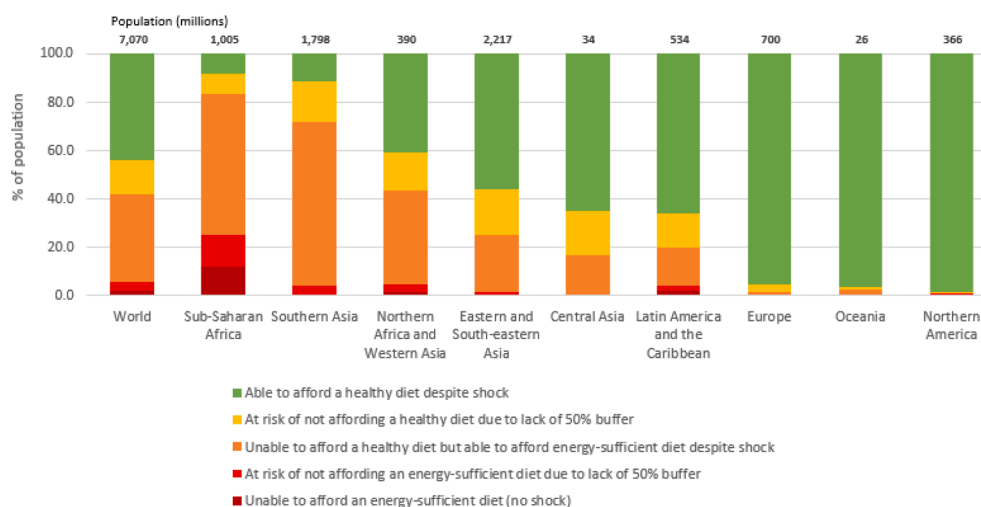


Fig. 3. Spectrum of diet affordability and vulnerability to a shock equivalent to a one-third loss in real income: percent of population, by region.

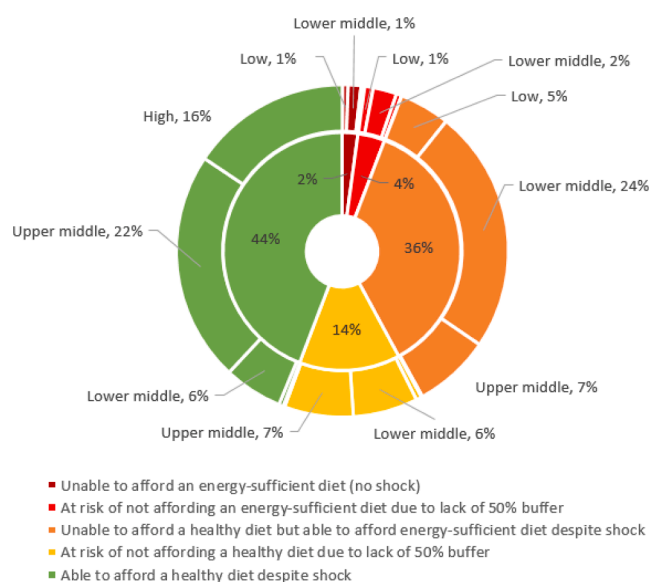


Fig. 4. Distribution of the global population along the spectrum of (un)affordability and risk of unaffordability of diets, by income level.

18–19% of the populations do not have enough income to buffer a shock reducing real incomes by one-third.

- (iv) Finally, and unsurprisingly, in Europe, Northern America and Oceania – mostly composed of high-income countries – more than 95% of the population can afford a healthy diet even in the face of a shock reducing real incomes by one-third.

Fig. 4 concludes this section by showing how the global population is distributed across the five categories of Figs. 2 and 3 – expressing (un)affordability and risk of unaffordability of diets. Lower-middle income countries are home to the majority of people who cannot afford an energy-sufficient diet or are at risk of not affording it. Although low income countries have a higher share of their population in these two categories than lower-middle income countries do (See Fig. 2), the latter are home to a bigger portion of the global population, and thus have a greater number of people in these two categories. Lower-middle income countries are also home to the majority of people who cannot afford a healthy diet. On the other hand, the population at risk of not affording a healthy diet is equally divided between lower- and upper-middle income countries.

Understanding the underlying drivers of unaffordability and the risk of unaffordability is central to this viewpoint, and is discussed in the following two sections. Based on the conceptual framework represented in Fig. 1, unaffordability and vulnerability to not affording a diet are shaped by three factors: the cost of the diet, the mean level of income and income inequality, or a combination of these.

3.2. Placing countries along the spectrum of diet affordability

This section illustrates where individual countries are placed along the spectrum of affordability of an energy-sufficient and a healthy diet. Doing so can help policymakers understand the underlying driver(s) of unaffordability and vulnerability – i.e. income inequality, poverty, the cost of foods that make up each diet, or a combination of these.

3.2.1. Unaffordability of energy-sufficient diets

Although countries ideally would aspire to provide a healthy diet to all, the first step in this direction is to make sure a country's population can at least meet basic dietary energy needs. Fig. 5 plots the share of each country's population unable to afford an energy-sufficient diet (vertical axis) against the Gross National Income (GNI) per capita

(horizontal axis) in 2019 for 58 of the 136 countries (results for all countries can be found in Table S1). The GNI per capita is intended as a measure of how likely country populations are – on average – to have an income buffer to face shocks. The different shades of markers flag the inability to afford under four different scenarios. The darkest red marker indicates the share of the population unable to afford an energy-sufficient diet even in the absence of a shock given 2019 income, and thus maps the dark red area in Fig. 1 representing the most vulnerable segment of a population. The remaining markers map the lighter red area in Fig. 1 – i.e. those able to afford an energy-sufficient diet but who, in the face of a shock, lack enough of a buffer of 11%, 25% and 50% relative to the affordability threshold, respectively, to ensure people can afford this diet after the shock.

The share of people who cannot afford a given diet or are at risk of not affording it depends, in part, on the cost of the diet relative to the mean income available for food. The lower this ratio, the cheaper the diet in relative terms and the higher the likelihood that a population is able to access it (ratios for each country are presented in Table S1 in the Supplemental Information), unless income inequality affects people's ability to afford it. Energy-sufficient diets are relatively cheap; therefore, if populations are still unable to afford one, this will likely be driven by an unequal distribution of income.

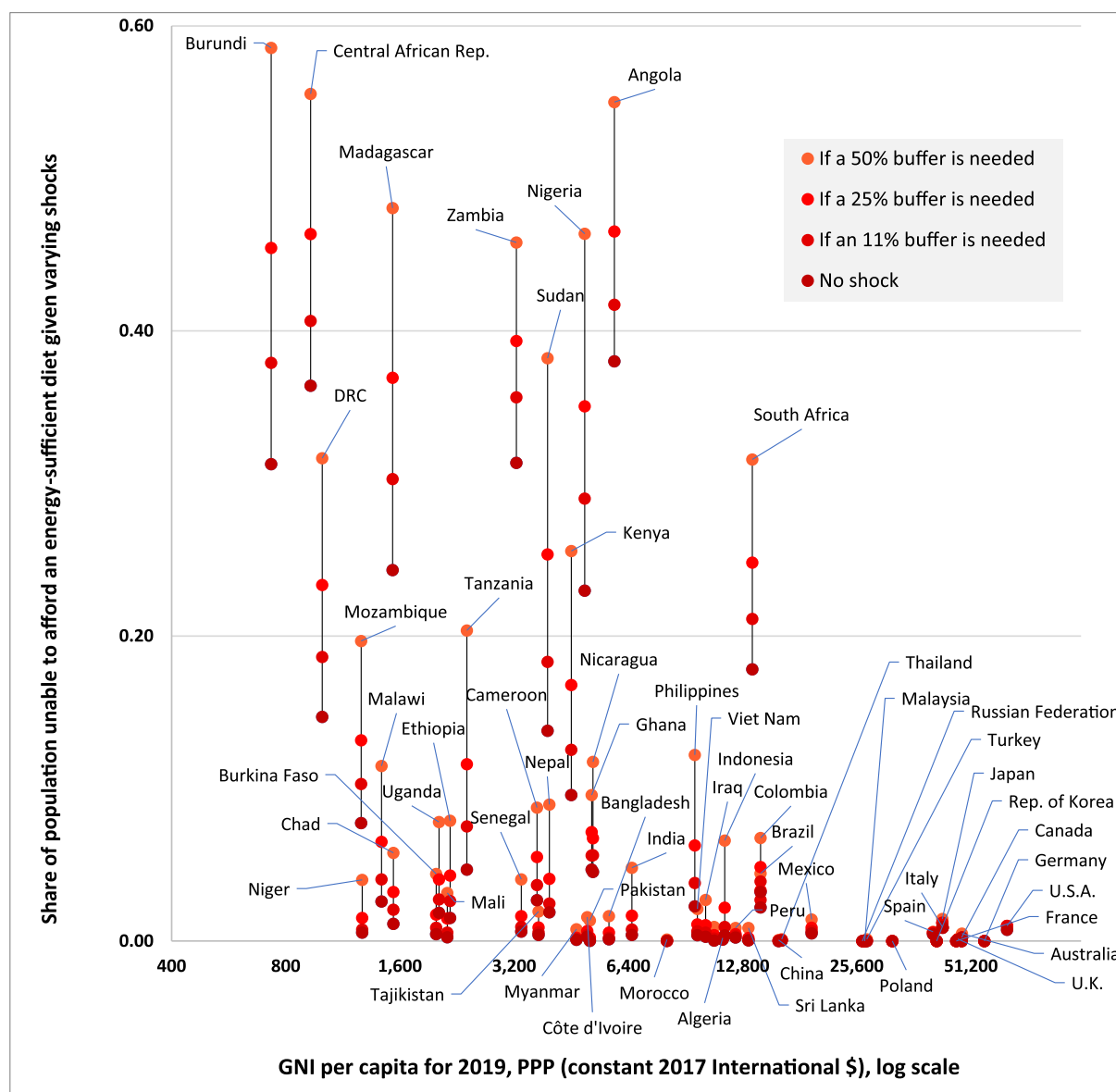
Looking at Fig. 5, economic access to an energy-sufficient diet tends to be lowest for countries with a very low GNI per capita, although not exclusively. For example, Burundi and the Central African Republic (GNI < 1 000) have approximately one-third of the population unable to afford, independently of shocks, while in the Democratic Republic of Congo (DRC), with an equally low GNI level, this share is much lower (15%). At the same time, countries with a considerably higher GNI than DRC, such as Angola, Nigeria, and South Africa, can have a higher share of people who are unable to afford the diet. This variation suggests that other factors beyond mean income level – namely income inequality and the relative cost of diets – may come into play.

A more illustrative example is that of Madagascar and Niger, two countries with very similar GNI levels, but where the level of unaffordability after a shock requiring a 50% income buffer (relative to the affordability threshold) goes from 4% in Niger to 48% in Madagascar. This variation is mostly driven by the cost of the energy-sufficient diet relative to mean income, which is two times higher in Madagascar, and partly driven by higher income inequality.^d Comparing Niger and Mozambique further exemplifies the role of income inequality: both countries have very similar relative diet costs and income levels; however, Mozambique is a more unequal country (Gini index = 54) than Niger (Gini index = 37). Consequently, Mozambique has 8% of its population unable to afford the diet, and a further 12% at risk, which is significantly higher than in Niger.

Unsurprisingly, high- and upper-middle income countries have a very small share of the population vulnerable to not affording an energy-sufficient diet, although exceptions exist. South Africa is a case in point, with up to 31% of its population being vulnerable to a one-third reduction in real income. Further, there can also be considerable variation. For example, Brazil and Colombia have a similar GNI per capita level as in South Africa but the diet is relatively much cheaper in these two countries, resulting in a considerably lower share of vulnerability (4% and 8%, respectively), though still worrisome for countries of this level of income.

The extent to which policymakers should focus on those unable to afford a diet under any circumstance, as opposed to those able to afford but vulnerable to shocks is central to this viewpoint. A differential approach may be needed in allocating resources to help those unable to afford and those at risk of unaffordability. Take the case of India and Indonesia, whereby only a very small share of the population cannot afford an energy-sufficient diet even in the face of a shock requiring a

^d See Table S1 for the cost of diets and Gini coefficient in each country.



Note: For 15 countries, GNI per capita for the year 2019 is not available and the year 2017 is used instead. Data and results on the full list of countries are available in Table S1 in the supplemental information.

Fig. 5. Economic access to an energy-sufficient diet with incremental shocks. Note: For 15 countries, GNI per capita for the year 2019 is not available and the year 2017 is used instead. Data and results on the full list of countries are available in Table S1 in the supplemental information.

25% income buffer. Yet, this share increases substantially if the shock requires a 50% buffer relative to the affordability threshold.

3.2.2. Unaffordability of healthy diets

As mentioned, the cost of a healthy diet is almost 5 times the cost of the energy-sufficient diet, and this pattern holds across all regions and country income groups. Referring back to our framework in Fig. 1, this means that countries where a considerable share of the population could not afford the cheaper energy-sufficient diet will inevitably have an even larger share who cannot afford a healthy diet. However, given the cost difference between the two diets, there will also be countries where the population does not have a problem affording energy-sufficient diets even when shocks unfold, but where a sizeable share cannot afford a healthy diet even without a shock.

Fig. 6 complements Fig. 5 by focusing on the economic access to healthy diets. The darkest marker in Fig. 6 maps the orange area and the two red areas in Fig. 1 – i.e. those unable to afford a healthy diet with

their 2019 income level. The lighter markers map the yellow area in the income distribution curve of Fig. 1 – i.e. those able to afford a healthy diet with their 2019 income level but who lack enough of a buffer to afford the diet following shocks of different magnitudes. Fig. 6 confirms that the level of affordability of a healthy diet is particularly low for countries with a sizeable share of the population being vulnerable to not affording an energy-sufficient diet. This is the case of Angola, DRC, Madagascar, Nigeria, South Africa and Sudan where more than 90% of the population cannot afford a healthy diet, independently of any shocks. Furthermore, a sizeable number of low-income countries that had a relatively small share of the population at risk of not affording an energy-sufficient diet, are found to be vulnerable to not affording a healthy diet, such as Burkina Faso, Ethiopia, Niger and Uganda. With the exception of South Africa, these are all countries where the ratio between the cost of a healthy diet and mean income available for food is above 1 (see Table S1). In these countries, increasing diet affordability will likely require increasing incomes, lowering the cost of a healthy

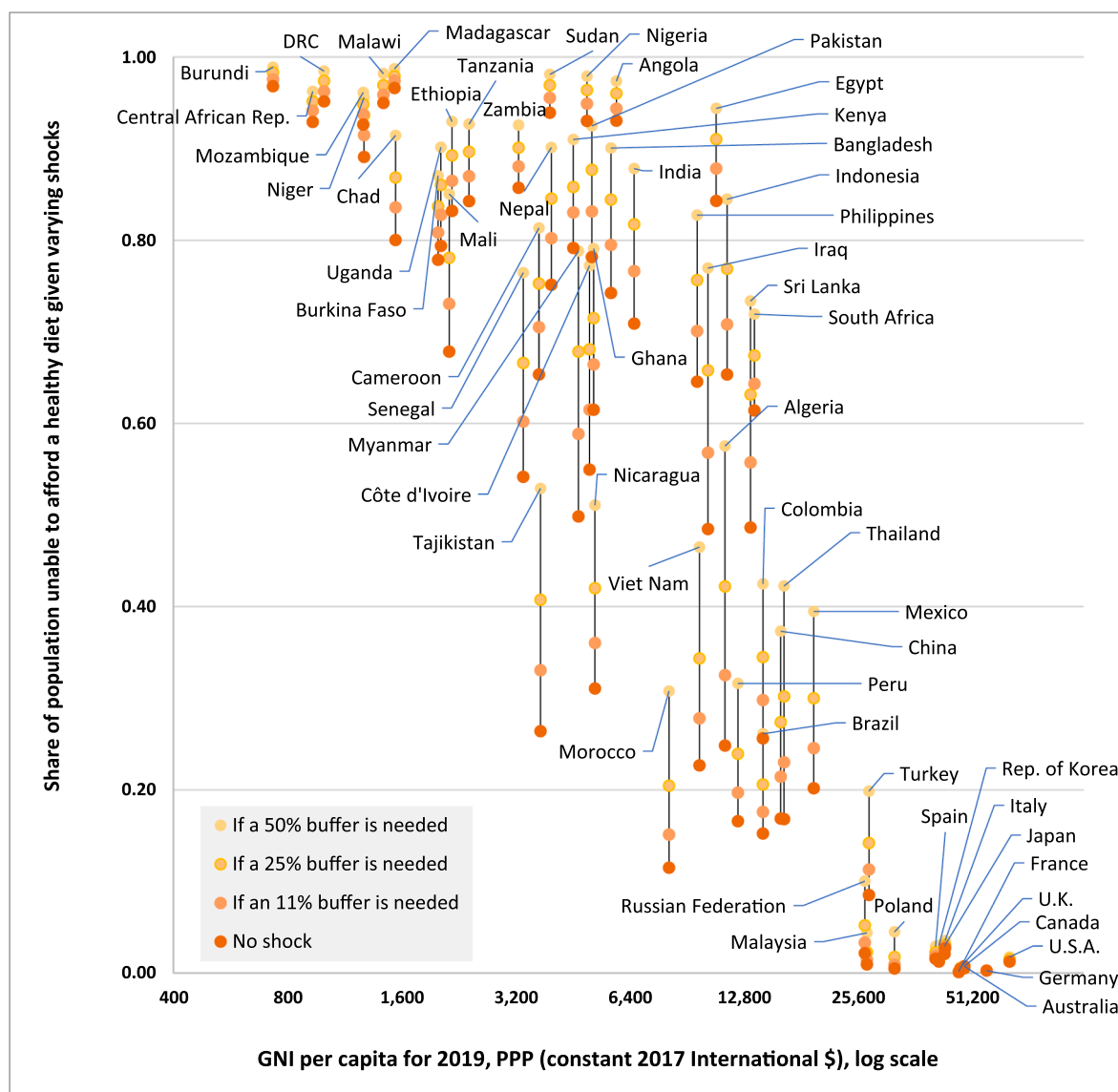


Fig. 6. Economic access to a healthy diet with incremental losses in real income for a subset of countries. Note: For 15 countries, GNI per capita for the year 2019 is not available and the year 2017 is used instead. Data and results on the full list of countries are available in Table S1 in the supplemental information.

diet, or a combination of both. Paradoxically, for these countries, if income were to be re-distributed uniformly, nobody would be able to afford a healthy diet.

As GNI per capita increases, the level of affordability can differ across countries with the same average income. **This is the case of Cameroon and Tajikistan: the latter has a much lower share who cannot afford a healthy diet (26% vs 65%) but a higher share at risk of unaffordability (27% vs 16%) despite a similar GNI per capita.** This can be partly attributed to higher inequality in Cameroon (Gini index equals 47 for Cameroon compared to 34 for Tajikistan), in addition to a cost of the diet relative to mean income that approaches unity (see Table S1). Another similar comparison can be made between the Philippines and Viet Nam, with the former having a higher share who cannot afford a healthy diet, but the latter still having a sizeable share of the population at risk of not affording it. Thus, depending on country context, a differential approach may be needed in allocating resources to help those unable to afford a

healthy diet and those at risk.

The distance between a country's four markers indicates the vulnerability of the population to different shocks. For example, the populations of countries like Morocco, Tajikistan, and Viet Nam seem to be fairly resistant to a shock requiring an income buffer of 11% relative to the affordability threshold, but are more than proportionally vulnerable to shocks requiring an income buffer of 25% and even more of 50%. Larger shocks are less likely, but would affect considerably larger swaths of the population in terms of their ability to afford a healthy diet. This is particularly the case among middle-income countries. On the other hand, countries with the highest GNI per capita (e.g. Germany, Republic of Korea, and the United States of America) are resilient to shocks that require even an income buffer of 50% or more, as populations can likely address their food needs by diverting a part of their sizeable non-food expenditures.

Fig. 6, combined with Fig. 5, can provide guidance on which diet to

Table 1

Regression results for predictors of share of population along the spectrum of vulnerability in access to diets.

	More vulnerable → Less vulnerable			
	Share who cannot afford energy-sufficient diet	Share at risk of not affording energy-sufficient diet	Share who cannot afford healthy diet but can afford energy-sufficient diet after shock	Share at risk of not affording healthy diet
Constant	-50.81*** (10.55)	-38.53*** (8.83)	-8.60*** (3.10)	-6.41 (4.53)
Gini	14.53*** (2.73)	12.23*** (2.278)	2.89*** (0.77)	1.76 (1.15)
Mean income	-1.93*** (.55)	-2.87*** (.46)	-1.67*** (0.17)	-1.98*** (.24)
Diet cost	0.97 (1.31)	2.42** (1.09)	3.57*** (.78)	5.82*** (1.14)
R-squared	0.3509	0.4988	0.6382	0.5376
Number of countries	136	136	136	136

Standard errors are reported in parenthesis. **, *** indicates significance at the 95%, and 99% level, respectively.

place the policy focus, particularly in middle- and low-income countries. For example, in Bangladesh, where access to an energy-sufficient diet is not an issue but where nearly 75% of the population is unable to afford a healthy diet, policymakers should focus on improving access to the latter. Conversely, in India and Indonesia the share of population unable to afford a healthy diet is slightly lower than that of Bangladesh, but up to 5% is at risk of not affording an energy-sufficient diet. The most problematic countries have more than 15% of the population without access to an energy-sufficient diet, or at risk of losing it. In such countries, some of which are affected by conflict, it is likely that broader development strategies are needed to lift the large segment of the population out of poverty and ensuring access to the energy-sufficient diet.

4. Policy entry points to reduce vulnerability to not affording energy-sufficient and healthy diets

Poverty, income inequality and the cost of diets appear to be critical underlying drivers of the observed numbers in Figs. 5 and 6. We will now decompose these effects in terms of their statistical significance and relative magnitude. Based on the conceptual framework in Fig. 1, combined with the data described in the Methods, we use an ordinary least-squares (OLS) model to decompose the predictors of vulnerability in affording a diet. As explained in Fig. 1, people may be vulnerable to not affording a given diet because they lack enough of an income buffer following a shock or, even more worrisome, because their disposable income is below the cost of the diet, preventing them from affording it even in the absence of a shock. Clearly, populations under the latter scenario are more vulnerable than those in the former. Understanding the predictors of these levels of vulnerability is at the centre of this viewpoint.

The predictors considered are a country's level of mean income per capita per day,^e the cost of a diet and the degree of inequality in the distribution of income measured by the Gini index. We transform logarithmically the variables to obtain a linear relationship in elasticities between the dependent variable and its predictors. Details of the underlying model are presented in Annex 2. The dependent variables are reported in the column headings of Table 1 following the spectrum presented in Fig. 1 – from most vulnerable (left) to least (right).

Findings in Table 1 indicate that:

- (i) Starting from the most vulnerable (first column), i.e. from those unable to afford an energy-sufficient diet even in the absence of a shock, income inequality is confirmed to have the strongest association with the share of population that cannot afford this basic diet. Mean income also reveals a strong (inverse) association. Conversely, diet cost is not statistically significant, which is to be expected because this diet is relatively cheap even in low-income countries.
- (ii) For those at risk of losing access to an energy-sufficient diet (second column), inequality still dominates although the magnitude of the association with mean income increases. Diet cost is now statistically significant, a reasonable result since we are now considering the case of a shock reducing real incomes by one-third (either through a price increase or income shortfall), meaning that differences in relative cost of the diet are amplified. This suggests that income inequality remains the main concern, in the long term, for policymakers interested in limiting the risk of unaffordability of energy-sufficient diets; however, price stabilization may play a more immediate role, especially if a shock affects staple food prices.
- (iii) For people who are not at risk of affording an energy-sufficient diet but who still cannot afford a healthy diet (third column), we find that income inequality, mean income, and diet cost are all statistically significant, with the expected sign, and with the same order of magnitude in terms of their association with the share of population in this category. This suggests that focusing on inequality is still desirable, but not necessarily the only option since increasing mean incomes or reducing the cost of the diet can also be effective.
- (iv) Finally, for those who are vulnerable to losing access to a healthy diet should a shock occur, we find that income inequality measured by the Gini coefficient is not significant, indicating that income redistribution would not help those people in this category unlike those in the more vulnerable categories. Increasing mean incomes and especially reducing the cost of the diet would be more effective.

These results hold also for smaller shocks that require less than a 50% income buffer. Table S2 presents results applying to shocks requiring an income buffer of 11% and 25% and we find the coefficients are similar in

^e Mean income per capita per day were retrieved from the World Bank Poverty and Inequality Platform (World Bank, 2022b).

significance, sign, and magnitude. The sensitivity analysis over this range of shocks indicates that for individuals struggling to afford an energy-sufficient diet, a one percent reduction in income inequality could provide a benefit 3 to 7 times that of a similar increase in mean income per capita. The robustness of these findings confirm that inequality plays the most important role in influencing accessibility to a basic energy-sufficient diet, while mean income and particularly diet cost are increasingly relevant for a healthy diet, especially in reducing vulnerability to shocks.

5. Policy discussion

Combining the “macro” dimensions of food security in [Timmer \(2000\)](#) with the results presented in this viewpoint, we conclude that the level of economic development (expressed by mean income), income inequality, and relative cost of diets are critical elements in determining access to different diets and the share of those vulnerable to not affording them in case of a shock. However, their relevance varies along the affordability spectrum presented in [Fig. 1](#). Addressing income inequality stands out when trying to ensure access to energy-sufficient diets, while mean income and cost of diets are increasingly relevant elements in reducing vulnerability to shocks in accessing a healthy diet.

Given the impact the COVID-19 pandemic and, more recently, the war in Ukraine, have had on incomes, income inequality and food prices at the global level, we envision three pathways to address the challenge of increasing inaccessibility to diets:

- Intervening along food supply chains to lower the cost of nutritious (and/or staple) foods and stabilize their prices
- Strengthening resilience to economic adversity through social protection programs and income stabilization
- Tackling poverty and structural inequalities

Pursuing these pathways will depend on where a country sits along the affordability spectrum, as well as on its fiscal and macroeconomic conditions ([FAO et al., 2021](#)). Several pathways can apply simultaneously and may complement and reinforce each other.

Starting from the most vulnerable populations where even a basic energy-sufficient diet is out of reach, or at risk of becoming so, tackling income inequality and poverty will be particularly important. Possible options for countering inequality are targeted cash transfers, prioritizing spending for long-term growth, and mobilizing tax revenue without hurting the poor ([World Bank, 2022a](#)). In Ethiopia, for example, where access to energy-sufficient diets can be improved, the Productivity Safety Net Programme provides cash or food transfers to chronically food-insecure households, along with financial and technical support, to help them build assets that can sustain them through future crises. This could be alongside longer-term structural efforts to improve economic access to healthy diets. For example, evidence is emerging that improving the allocation of the existing public budget for agriculture in Ethiopia, with a view to accelerate agricultural transformation, would increase value for public money, not only in terms of agrifood output growth, job creation and poverty reduction, but also in terms of increasing the affordability of healthy diets for more than 2 million people ([Sánchez and Cicowiez, 2022](#)). In the long-term, policies may also be needed to address situations of conflict, protracted crises, low agricultural productivity and limited agrifood infrastructure to improve access, including to healthy diets.

Social protection measures that help stabilize incomes will also be key, especially to ensure continuous access to energy-sufficient diets after a shock. In fact, many countries have moved in this direction due to rising inflation. [Gentilini et al. \(2022a\)](#) report a 460% increase in the number of social protection measures between April and December 2022, summing to 1 016 responses across 170 economies for a social protection spending of 710 billion USD. This comes after a dramatic expansion of social protection measures in connection to COVID-19,

which went from 103 in March 2020 to 3 856 in February 2022, summing up to 3 trillion USD in expenditures ([Gentilini et al., 2022b](#)). COVID-19 response measures mostly included cash transfers, in-kind food and voucher schemes, and school feeding. The social protection measures in connection to the recent increase in inflation instead included a broader set of measures: subsidies, social assistance, tax measures, trade-related measures, social insurance, and labor market programs. These measures were in most cases aimed at maintaining real incomes that were being negatively affected, and are also of utmost importance to ensure access to healthy diets during times of crisis. It is difficult to identify who were the beneficiaries of these programs and how they were targeted. The data reported indicates that these programs, in relation to affordability of diets, likely fell short in low-income countries – where, on average, 8% of the population was reached by COVID-related cash transfer measures – but where we estimated that approximately 20% of the population either could not afford an energy-sufficient diet or was at risk given the income shortfalls experienced in that period. At the other extreme, in high-income countries where very few were at risk, cash transfers reached on average 44% of the population.

The fact that income support measures were not enough to ensure access to even a basic energy-sufficient diet to all again supports the idea that social protection cannot be the sole solution at addressing diet unaffordability. In more extreme cases, traditional price stabilization policies – such as used for rice in Asia – could be useful to improve affordability to energy-sufficient diets, at least in the short term. Indeed, this approach was pursued by 37 countries who restricted export of specific food items in 2022 – e.g. for basic staples in Ghana, India, Mali and Uganda, chicken in Malaysia, and sugar in Pakistan ([Gentilini et al., 2022a](#)). For the many people around the world who are at risk of losing access to an energy-sufficient diet, price stabilization of staples on which they rely heavily can play an important role when a disruption may require a country to focus on ensuring that people can afford basic staples.

Given the diversity of commodities composing healthy diets, price stabilization will be more challenging relative to what was done for staple crops in the recent past. Yet, mixed policies that ensure access to starchy staples while attempting to diversify towards other relatively cheaper components of a healthy diet (e.g. oils and plant-based proteins) can be an option. Countries like Egypt and Cameroon have already put trade restriction on vegetable oils and plant-based proteins alongside staples like flour and cereals ([Gentilini et al., 2022a](#)).

Further, policymakers can also intervene along food supply chains to lower the cost of nutritious foods that make up a healthy diet. Policies can include investments from production to consumption aimed at realizing efficiency gains, stimulating diversification of food production and nutritious foods, investing in innovation, research and extension to raise productivity, and cutting food loss and waste.

6. Conclusions

This viewpoint article contributes to a more comprehensive picture of the resilience of national agrifood systems in terms of economic access to healthy diets and energy-sufficient diets. It is intended to inform countries on the vulnerability of their populations, and provide guidance on how to absorb impacts of shocks on access to food. The main findings indicate that there is high heterogeneity among countries regarding the level of economic access to energy-sufficient and to healthy diets. There is a large share of the population that is vulnerable to losing access to diets when confronted with a shock that affects their purchasing power. Such vulnerability is mostly due to income inequality, in the case of an energy-sufficient diet, and to a confluence of limited incomes and a relatively high diet cost when it comes to a healthy diet.

We identified broad country profiles for access to energy-sufficient and healthy diets, respectively. At one extreme, we have countries –

especially low-income in Asia and sub-Saharan Africa – where more than 80% of the population cannot access a healthy diet and are in dire need of greater affordability, independently of shocks. At the other, we have high-income countries with limited affordability issues. Many households in middle-income countries are also vulnerable given that many of their populations are at risk of not being able to afford a healthy diet should a shock to real incomes occur.

Policy focus and required investments will depend on where a country sits on this spectrum. Low-income countries may place more emphasis on addressing structural issues such as overall income levels, income inequality, and food system infrastructure as well as ensuring agricultural transformation is accelerated to increase agricultural productivity and incomes, while in other countries focus may be more on risk management through income stabilization mechanisms. In all cases, social protection programmes and how they are designed will be key. For most countries, investments and policy interventions will also be needed to ensure affordability in general, and following disruptions. Considering national fiscal constraints and macroeconomic environments, a political economy of resilience is also required to assist the marginalized.

Particularly worrisome is the fact that numerous countries – mostly low-income in sub-Saharan Africa – have many people unable to even afford a basic energy-sufficient diet meeting caloric needs if a significant shock occurs. In these cases, raising the purchasing power of the poorest through social protection mechanisms is a key short-term option. When designed to be gender- and nutrition-sensitive and responsive to multiple risks and shocks, these programmes can provide timely support to at-risk and crisis-prone populations, as was observed during the COVID-19 pandemic. Cash transfers have been the most common. In the long-term, policies are needed to address situations of conflict, protracted crises, low agricultural productivity and limited agrifood infrastructure to improve access to healthy diets. Policies aimed at reducing poverty and income inequality while enhancing employment and income-generating activities, are key to raise the affordability of diets (FAO et al., 2020). The results presented here highlight how these are also key in ensuring resilient economic access to diets when shocks occur. As pointed out by Timmer (2000), when discussing the macro dimensions of food security, these are often structural issues that invest the broader economy beyond the agrifood system.

Several caveats apply to the results presented here. First, our analysis is based exclusively on costs of diets and income levels and their distribution, and does not take into account any government programmes that are already in place, such as social protection schemes. Therefore, the policy guidance we provide may already be implemented in part, or in its entirety, by policymakers in some countries. Second, by relying on national averages, we do not consider how within-country variability of the costs of diets may affect the reported results. Third, the buffer considered to be needed after a shock best represents the time period immediately following the shock, since it does not consider adjustments that may take place to compensate for an income shortfall, general equilibrium effects of the shock, or what policies are implemented.

In summary, improving access to energy-sufficient and to healthy diets for all requires significant change within agrifood systems and beyond. Given unique country contexts and different starting points for the various transformative processes that need to happen, this analysis has presented how countries will need to put different emphasis on long-term development strategies to improve incomes versus managing risk in connection to specific shocks, and has laid out the policy implications of these differences across countries.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodpol.2023.102470>.

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